CHICAGO BUILDING PERMITS

Investment Opportunity Assessment

Real Estate Developers & Investment Firms

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Today's Analysis

01 The Data

Chicago building permits (2024–present) + neighborhood socioeconomics

02 Where Capital Flows

Geographic concentration across 77 community areas

03 Project Mix

Renovation-heavy market: 82% renovations vs 18% new construction

04 Demographics & Returns

How income levels drive investment activity

05 Predictive Models

Linear & logistic regression to predict project cost and size

06 Recommendations

3-tier investment strategy by capital level and risk tolerance

The Data

PRIMARY DATASET

Chicago Building Permits

Source:	City of Chicago Socrata API
Date Range:	January 2024 – Present
Records:	~60,000+ permits
Key Fields:	Cost, permit type, community area, date

SECONDARY DATASET

Chicago Socioeconomic Indicators

Source:	City of Chicago Socrata API
Coverage:	All 77 community areas
Key Fields:	Per capita income, poverty rate, unemployment
Purpose:	Neighborhood investment context

Where Capital Flows

$3.9B

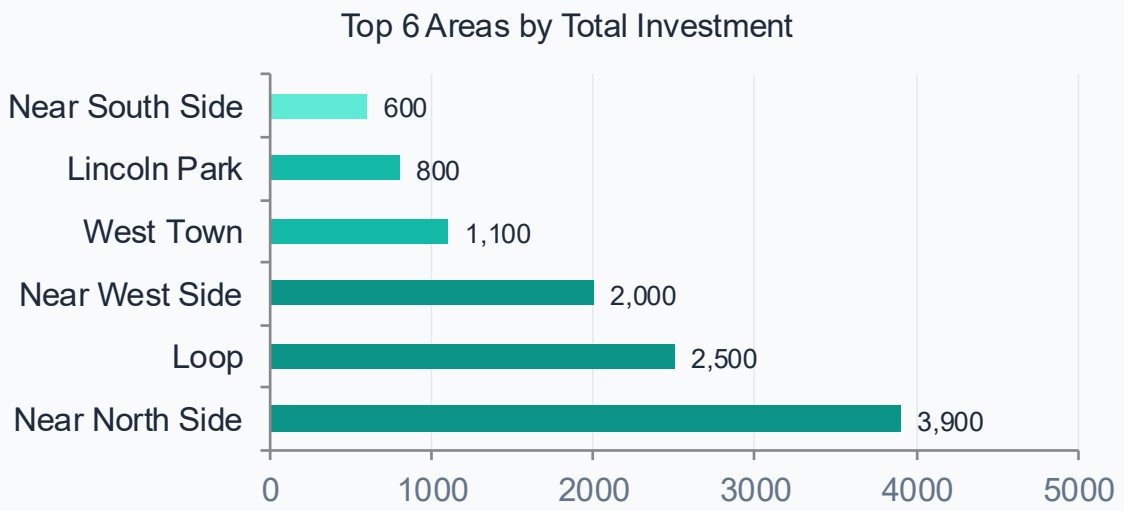
Near North Side
(#1 by investment)

r = 0.94

Income ↔ Investment
correlation

Top 5

Areas = >50% of
all investment



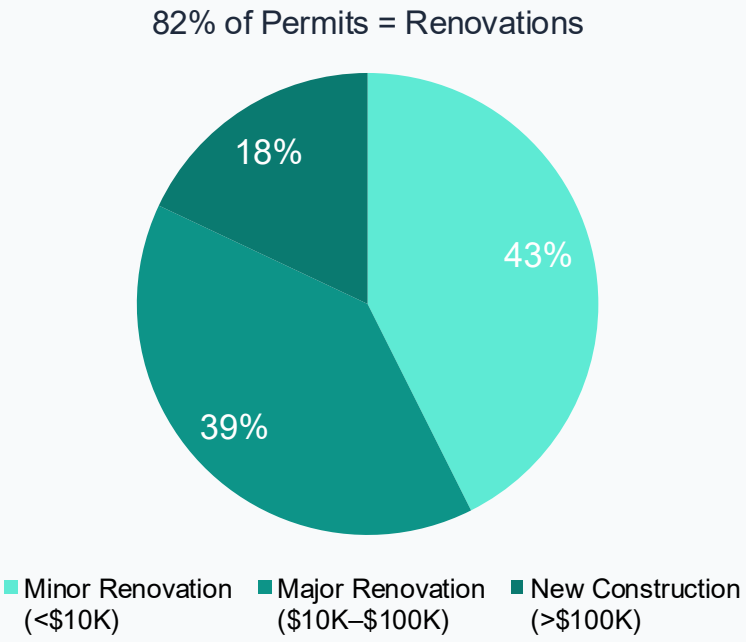
Key Insight

"Money follows money"

High-income neighborhoods dominate investment. North Side & Downtown capture the vast majority of development capital.

Gentrification signal: 8–12 lower-income areas show above-median construction activity.

Project Mix & Capital Requirements



Strategy	Median Cost	% of $ Volume
New Construction	$300K	95.6%
Major Renovation	$24K	4.0%
Minor Renovation	$3K	0.5%

Capital-Intensive

>$500K

New construction in Tier 1 markets (North Side, Loop)

Value-Add

$100K–$500K

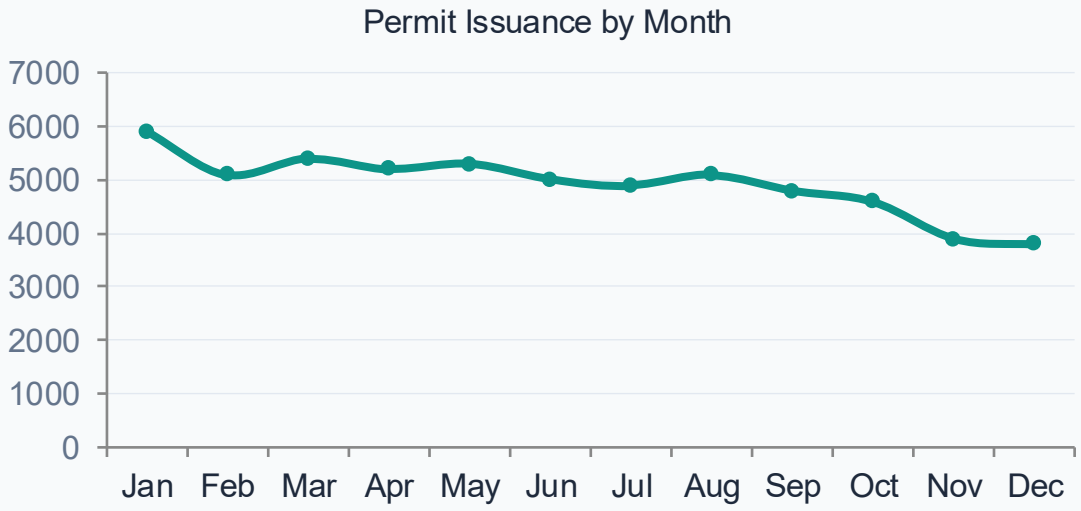
Major renovations in Tier 1 & 2 markets

Volume-Based

<$100K

High-volume minor renos on South/West Side

EDA Highlights



Seasonal Insights

Peak: Jan–May
Contractors plan spring/summer work

Trough: Nov–Dec
35–40% fewer permits issued

Action: Submit permits Nov–Feb for faster processing

Q4 Income Areas

60%+ of all investment

Downtown/Loop

Dwarfs all other regions

Fee Ratio

Consistent across income tiers

Predictive Models

MODEL 1 | LINEAR REGRESSION

Predicts project cost (log scale)

R^2	Explains variance in log-cost
RMSE	Root mean squared error
MAE	Mean absolute error

Top predictors:

New Construction flag, per capita income, poverty rate

MODEL 2 | LOGISTIC REGRESSION

Classifies major projects (≥ $100K threshold)

Acc.	Accuracy on held-out test set
Prec.	Of predicted major, how many correct
Recall	Of actual major, how many caught
F1	Harmonic mean of precision & recall

Business use:

Screen neighborhoods for high-probability major development before committing capital

Investment Recommendations

TIER 1

Premium Markets

Score > 0.45

Areas	Near North Side, Lincoln Park, Loop, Lake View
Strategy	Major construction & new development
Capital	> $500K per project
Risk	Low

TIER 2

Emerging Opportunities

Score 0.40–0.45

Areas	Near South Side, Edison Park, Beverly
Strategy	Value-add renovations
Capital	$100K – $500K
Risk	Medium

TIER 3

Volume Markets

Score < 0.40

Areas	South Side & West Side
Strategy	High-volume minor renovations
Capital	< $100K per project
Risk	Higher / Operational

Conclusions & Next Steps

Key Takeaways

- Money follows money — r=0.94 income↔investment correlation
- 82% of permits are renovations; new construction drives 95.6% of dollars
- 8–12 gentrification-stage areas offer the best risk-adjusted upside
- Seasonality is real — submit permits Nov–Feb for faster approvals
- Investment score framework enables systematic, data-driven screening

Future Work

- 5–10 year historical permit trends
- Crime & school quality data integration
- Transit proximity analysis
- ML neighborhood forecasting models
- Validate investment score weights with actual ROI data

Thank you · Questions?